# Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 7 · SLO 2 — Solved Worksheet

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## Unit â€“ 3 : Session â€“ 7 : SLO â€“ 2 | Implementation of Stack Using Linked List â€” Code

### ■ I. Node Structure

• struct node { int data; struct node *next; };
• struct node *top = NULL;

### ■ II. Push Operation

• void push(int val) {
• struct node *new_node = (struct node *)malloc(sizeof(struct node));
• new_node->data = val;
• new_node->next = top;
• top = new_node;
• }

### ■ III. Pop Operation

• int pop() {
• if (top == NULL) { printf("UNDERFLOW\n"); return -1; }
• struct node *ptr = top;
• int val = ptr->data;
• top = top->next;
• free(ptr);
• return val;
• }

### ■ IV. Peek Operation

• int peek() {
• if (top == NULL) { printf("STACK IS EMPTY\n"); return -1; }
• return top->data;
• }